

Hamza Rauf

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ACHIEVEMENTS

- Collaborated with a **Stanford World's Top 2% Scientist** on applying advanced machine learning to solve complex problems in computational fluid dynamics.
- Published **six peer-reviewed** journal articles spanning resource-efficient language models, scientific machine learning, computational fluid dynamics and cybersecurity.
- Published a **first-author** paper on efficient, on-device AI in the **Q1-ranked journal** *Complex & Intelligent Systems*.
- Published a co-authored paper on ML surrogate modeling for magnetized second-grade nanofluid flow in the **Q1-ranked journal** *Engineering Applications of Artificial Intelligence*.
- **Co-authored** a 2026 paper in the **Q2 journal** *Asia-Pacific Journal of Chemical Engineering*, comparing Morlet wavelet neural networks trained with Levenberg-Marquardt and Adam.
- Awarded a **fully-funded merit scholarship** for my bachelor by ranking among the top 100 students nationwide, recognizing exceptional academic potential.
- Obtained an exceptional **GRE score of 331/340** (Quant: 165, Verbal: 166), placing in the top percentile and demonstrating elite analytical and reasoning skills.
- Selected as a **Section Leader** for **Stanford University's** global *Code in Place* program, mentoring international students in Python programming over a 6-week course.
- Member of a **Harvard CS50x** Puzzle Day 2024 team that solved 9 out of 9 puzzles.
- Maintained consistent academic excellence by graduating in the **top 5%** of the Computer Science class at Sukkur IBA University.
- Recognized as the **first student** in the academic batch to secure a professional job offer prior to graduation, highlighting high industry demand for my skills.
- Over **5 years** of **professional** experience in the tech industry, currently working as a Senior Software Engineer at one of the Pakistan's top technology companies.
- Spearheaded the end-to-end development of the **HelloGov AI** platform, an AI-powered solution for U.S. government document processing that has successfully scaled to over 50,000 applications annually for more than 10,000 premium users.
- Recognized as **Employee of the Year** for two consecutive years (2023, 2024) at Coding Crafts for outstanding contributions and leadership in AI development.
- Secured a competitive prize at the **NETSOL AI Hackathon** by rapidly designing and implementing a high-impact AI solution under strict time constraints.

EDUCATION

Bachelor | Computer Science

Sukkur IBA University, Pakistan | Feb 2018 - Feb 2022

- CGPA : 3.31 / **85%**
- **Distinction** in 24 courses.
- **Focus:** Core coursework in Research Methodology, Artificial Intelligence, Machine learning, Computer Vision, Big Data, and Algorithms.

PEER-REVIEWED PUBLICATIONS

Efficient Multilingual Spam Detection on Resource-Constrained Devices: A Comparative Analysis of QLoRA Fine-Tuning of Gemma 3, Qwen 3, and Llama 3.2 Models.

Journal : *Complex & Intelligent Systems (Q1 Journal)*

Impact Factor : 5

Date : Nov 18th 2025

Authors : Hamza Rauf, Umair Khan, Jomana A. Bashatah, Aurang Zaib

Link: [Click here to view](#) 

This research pioneers a novel, privacy-preserving framework to address multilingual spam detection on resource-constrained devices. The methodology overcomes data scarcity by synthetically generating a balanced, 21-language dataset and employs Quantized Low-Rank Adaptation (QLoRA) to efficiently fine-tune compact LLMs. A rigorous comparative analysis identified the Gemma 3-4B model as the top performer on, achieving a state-of-the-art 90% accuracy and a 22-point improvement over its baseline. This work provides a validated blueprint for deploying powerful, privacy-preserving AI classifiers in environments with limited computational resources.

Comparative Analysis of Advanced Machine Learning Models for Second-Grade Liquid via Stimulating CoFe₂O₄ Nano Particles over a Convective Moving Plate.

Journal : *Engineering Applications of Artificial Intelligence (Q1 Journal)*

Impact Factor : 8.0

Date : Dec 19th 2025

Authors: Umair Khan, Hamza Rauf, Aurang Zaib, Abeer M.Alotaibi

Link: [Click here to view](#) 

This research investigated the complex, time-dependent flow of a magnetized second-grade nanofluid, a system with significant industrial applications. To overcome the computational expense of traditional numerical solvers, this work pioneered a comparative analysis of advanced machine learning surrogates. Our findings quantitatively demonstrated that Gaussian Process Regression (GPR) is a formidable high-fidelity tool, achieving near-perfect predictive accuracy ($R^2=1.0$) on par with the ANN-LM benchmark, while offering the invaluable capability of uncertainty quantification for real-world engineering design.

Residual Strength Prediction of Chemically Reactive Two-Phase Nanofluid Flow in a Bingham-Papanastasiou Rheological Theory Using a Morlet-Based Wavelet Neural Network Approach.

Journal : *Asia-Pacific Journal of Chemical Engineering (Q2 Journal)*

Impact Factor : 1.7

Date : Aug 9th 2026

Authors : Umair Khan, Hamza Rauf, Aurang Zaib, J. K. Madhukesh

Link: [Click here to view](#) 

This research examines chemically reactive, two-phase Bingham-Papanastasiou nanofluid flow under magnetic effects, velocity slip, and zero-mass flux conditions. Numerical solutions generated with MATLAB bvp4c were used to train Morlet wavelet neural networks with Levenberg-Marquardt and Adam optimization. The Levenberg-Marquardt model closely matched the numerical results and produced lower prediction errors than the Adam-trained model. The findings support Morlet wavelet neural networks as efficient surrogate models for predicting complex nanofluid behavior.

Using Generative AI for Simulating Cyber Security Attacks and Defense Mechanism: A new Approach to AI-driven Cyber Threat Modeling.

Journal : *Spectrum of Engineering Sciences*

Date : Mar 22nd 2024

Authors : Hamza Rauf, Syed Imran Hussain Shah, Taib Ali, Hassam Gul, Maria Soomro

Link : [Click here to view](#) 

This research introduces a proactive framework for AI-driven cyber threat modeling, leveraging Generative AI models like GANs and VAEs to simulate diverse and realistic attack scenarios. The study demonstrates how these simulations, which can emulate novel and advanced persistent threats, enable organizations to rigorously test and enhance their existing defense mechanisms. This approach facilitates a critical shift from a traditional reactive security posture to a dynamic and proactive strategy, better preparing systems against the evolving complexity of modern cyber threats.

Sentiment Analysis of Twitter for Electoral Process: A Systematic Literature Review.

Journal : *Spectrum of Engineering Sciences*

Date: Feb 22nd 2025

Authors : Hamza Rauf, Naveed Ullah Khan, Muhammad Rizwan Khaliq

Link : [Click here to view](#) 

This systematic literature review analyzes the application of Twitter sentiment analysis for predicting electoral processes and outcomes. The research methodically assesses key studies published between 2015 and 2020, focusing on critical aspects such as data collection techniques, preprocessing methods, and the algorithms employed. By synthesizing findings from high-impact databases, the work identifies significant gaps in the existing literature and concludes by proposing valuable future research directions for academics in this field.

Blockchain for Supply Chain Optimization in Industrial Engineering: Enhancing Transparency, Security, and Efficiency in Manufacturing Networks.

Journal : *Spectrum of Engineering Sciences*

Date : Feb 21st 2025

Authors : Syed Sheraz Ul Hasan Mohani, Syed Tauqeer Ali Shah, Abdul Waheed, Hamza Rauf

Link : [Click here to view](#) 

Co-authored a study investigating the transformative impact of blockchain technology on industrial supply chain optimization. Our research provided empirical evidence that blockchain integration significantly enhances transparency, security, and operational efficiency while reducing fraud. The work also identified technological readiness and robust IT infrastructure as critical success factors for implementation, offering a clear framework for organizations seeking to leverage blockchain for a competitive advantage.

ONGOING RESEARCH

Numerical analysis of radiative flow in a two-phase Bingham Papanastasiou Rheological nanofluid model: A comparative study using a Morlet-Based Wavelet Neural Network Optimization. (In Review)

Authors : Hamza Rauf, Umair Khan, Anuar Ishak

Investigates two-phase radiative nanofluid transport under a zero mass flux boundary condition to ensure physically consistent nanoparticle behavior. The similarity transformed nonlinear system is solved using MATLAB bvp4c to generate high fidelity benchmark data for skin friction coefficient and Nusselt number. Develops a Morlet-based Wavelet Neural Network surrogate and compares Levenberg Marquardt and Adam optimization strategies, showing LM achieves near-exact accuracy and enables fast parametric evaluation for thermo-fluidic design and optimization.

Machine Learning-Based Comparative Modeling of Buoyancy-Driven Non-Newtonian (Micropolar) Hybrid Nanofluid Flow. (In Review)

Authors : *Hamza Rauf, Umair Khan, Aurang Zaib, J.K. Madhukesh, Anuar Ishak*

Contributed to a research study applying advanced machine learning models to predict the complex behavior of non-Newtonian hybrid nanofluids, relevant to blood flow dynamics. The work involved a comparative analysis that identified an Artificial Neural Network trained with the Levenberg-Marquardt algorithm (ANN-LM) as the superior predictive tool. This model achieved near-perfect accuracy ($R^2=1.0$) and outperformed other techniques by reducing prediction error by over 90%, establishing a highly robust surrogate model for complex computational fluid dynamics.

Narrative Review – Large Language Models in Healthcare: A Taxonomy of Applications and Safety Considerations (In Preparation)

Developing a structured narrative review of LLM applications in healthcare, proposing a taxonomy that organizes systems by task, data modality, deployment setting, and model type. The work synthesizes current technical approaches (prompting, RAG, fine-tuning, multimodal models) and critically analyzes evaluation, safety, and fairness challenges for deploying LLMs in high-stakes clinical environments.

Embodied Emotion Classification via Small LLMs Using Prompting Strategies. (In Preparation)

Investigating fine-grained embodied emotion classification with **small LLMs** (e.g. **Gemma-3 4B** and similar models) by systematically benchmarking **zero-shot**, **k-shot**, **Chain-of-Thought**, and **Chain-of-Verification** prompting. The study evaluates macro/per-class F1 across Ekman's six emotions and highlights accuracy, efficiency tradeoffs to identify prompt designs that enable strong emotion recognition performance under limited compute.

THESIS

Roman Urdu and English Spam SMS Filtering/Detection | Sukkur IBA University

Date : *Jan 31st 2022*

Authors: *Hamza Rauf, Muhammad Noor Gohar*

Link: [Click here to view](#) 

Developed a comprehensive, multilingual spam filtering system to address the challenge of unwanted SMS in English, Roman Urdu, and Actual Urdu. The project involved creating novel datasets for under-resourced languages and implementing machine learning classifiers, with the Naive Bayes model achieving over 98% accuracy. The final solution was successfully deployed as a functional Android application, supported by a Flask-based backend for real-time language detection and classification.

RESEARCH EXPERIENCE

Research Assistant | Dr. Umair Khan (Stanford World's Top 2% Scientist)

Jun 1st 2024 - Present | Sakarya University, Turkey

- Co-authored three research papers (under review for journals like Journal of Molecular Liquids), applying advanced machine learning to solve challenges in computational fluid dynamics and multilingual NLP under the mentorship of Dr. Umair Khan.
- Developed high-fidelity Artificial Neural Network surrogate models to predict complex fluid dynamics, achieving a near-perfect coefficient of determination and reducing prediction error by over 90% compared to other techniques.

- Pioneered a privacy-preserving framework for efficient on-device AI by creating a novel 21-language synthetic dataset and employing Quantized Low-Rank Adaptation (QLoRA) to fine-tune compact Large Language Models.
- Achieved accuracy of 90% in multilingual spam classification with a QLoRA-tuned Gemma 3-4B model, representing a 22 percentage point performance uplift over its baseline and validating its viability for resource-constrained devices.

INDUSTRY EXPERIENCE

Senior Software Engineer | Coding Crafts

Nov 1st 2022 - Present | Pakistan / United States (Remote)

- Spearheaded the full-stack development and scaling of the HelloGov AI platform, an AI-driven service now serving 10,000+ monthly users in the U.S.
- Improved model performance by implementing a RAG system (40% accuracy gain) and applying Parameter-Efficient Fine-Tuning (LoRA) to Llama 3 (26% accuracy gain).
- Designed and deployed a GenAI agent-building platform for DataStax, featuring a drag-and-drop interface that drove a 35% uplift in user engagement.
- Enhanced team productivity by 20% through mentorship of 3 junior engineers, focusing on advanced system architecture, code quality, and modern DevOps practices.

Software Engineer | Valyrian System

Jan 1st 2020 - Oct 31st 2022 | Pakistan / United States (Remote)

- Architected and migrated a monolithic application to a high-availability microservices platform using Docker and Kubernetes, cutting API latency by 60% under peak load.
- Engineered and deployed a real-time network anomaly detection model with unsupervised learning, successfully identifying and mitigating over 95% of security threats in live traffic.
- Spearheaded a legacy modernization effort for the Yoga Joint platform by designing a new cloud-native system, which resulted in a 40% improvement in overall performance.
- Built a comprehensive automated testing suite from the ground up, increasing code coverage to 90% and reducing critical production bugs by 75%.

TEACHING EXPERIENCE

Section Leader – Python Programming (Stanford Code in Place)

Apr 22nd, 2024 - May 31st, 2024

- Led a small-group section for an introductory Python course, covering core concepts such as control flow, data structures, and functions.
- Provided mentoring, code reviews, and debugging support to help students complete weekly assignments and a final project.

Cloud Instructor (DevOps & Cloud Engineering at CloudSchool)

Feb 1st, 2025 - Apr 31st, 2025

- Taught modern DevOps practices and their implementation across major cloud providers (AWS, Azure, GCP).
- Designed and guided a mock production project where students implemented CI/CD, containerization, and cloud-native deployment pipelines.

INTERNATIONAL CONFERENCES

- Actively participated in international conferences at Sukkur IBA University, including the International Conference on Computing, Mathematics & Engineering Technologies (2020, 2022, 2024).
- Attended the "AI Essentials for Healthcare Leaders" conference at FAST-NUCES, 2025, gaining practical insights into AI applications for diagnostics, data management for the responsible and ethical adoption of AI in healthcare.

SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, SQL, Bash
- **AI / Machine Learning Frameworks & Tooling:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face (Transformers, Datasets, AutoTrain), Axolotl, OpenCV, YOLO, MATLAB, RapidMiner,
- **Generative AI:** Large Language Models (LLMs), Vision-Language Models (VLMs/VLLMs), Fine-tuning (full & PEFT/LoRA), Quantization (4-bit/8-bit), Retrieval-Augmented Generation (RAG), Reasoning fine-tuning, AI Agents, LangChain, Langgraph,
- **Web, API & Backend Development:** React.js, Next.js, Node.js, Flask, Django, FastAPI, REST APIs, HTML/CSS
- **Cloud, Systems & DevOps:** AWS, Azure, GCP, Docker, Kubernetes, Linux, Git, CI/CD, Kafka, RabbitMQ, Grafana, ELK Stack
- **Databases & Data Engineering:** SQL & NoSQL Databases, Vector Databases, Redis, PySpark, ETL Pipelines
- **Research & Academic Skills:** Systematic Literature Reviews, Qualitative & Quantitative Data Analysis, SPSS, ATLAS.ti, EndNote, LaTeX, Scientific Writing, Manuscript Preparation, Data Visualization for Publication

PROJECTS

Fine-Tuning a Multimodal Model for Medical Image Analysis:

This project demonstrated the specialization of a large multimodal model (Llama 3 Vision 11B) for advanced radiological analysis. Leveraging Parameter-Efficient Fine-Tuning (PEFT) with 4-bit quantization and LoRA, the model was efficiently adapted on a curated dataset of medical images. The successfully fine-tuned model is capable of generating accurate, context-aware diagnostic descriptions, showcasing a viable pathway for creating specialized clinical AI tools from general purpose foundation models.

Reinforcement Learning Alignment of Qwen3 using GRPO:

This project specialized the Qwen3-4B base model for complex mathematical reasoning by implementing a Generative Reinforcement Programming Optimization (GRPO) pipeline. The methodology involved a novel two-stage process: first, a preliminary fine-tuning to teach the model a structured, "chain-of-thought" output format, followed by GRPO training guided by custom reward functions that evaluated both logical coherence and final answer accuracy. This successfully aligned the model's outputs with correct problem-solving pathways, demonstrating a powerful method for instilling complex reasoning capabilities in compact LLMs.

Re-Agent: Verification-First LLM Agent for Binary Reverse Engineering

Built Reverse Engineering-Agent, a verification-first LLM agent with 20 MCP tools for binary analysis, symbolic execution, and SMT-based validation. Integrated angr, Claripy, and Z3 with an evidence ledger that separates proven results, observations, and unverified claims. Evaluated the system on four Mach-O x86-64 binaries covering code obfuscation, a custom virtual machine, and prompt injection. It passed all 36 benchmark checks and 38 offline tests, with recovered keys confirmed through real binary execution.

Development of a Transformer-Based Language Model from Scratch:

This project demonstrated the end-to-end development of a generative language model from first principles, building a 60-million-parameter, GPT-style transformer in PyTorch. The work involved implementing all core components, including causal self attention, layer normalization, and a high-performance training loop with mixed-precision. Successfully trained on the TinyStories dataset, the final model is capable of generating coherent, contextually relevant narratives, showcasing a deep, practical understanding of modern LLM architecture.

High-Efficiency Face Mask Detector for Embedded Systems:

This project involved the development and deployment of a high-efficiency face mask detector designed for real-time performance on resource-constrained devices. The solution employed a two-stage pipeline, first using a lightweight Haar Cascade classifier for rapid face localization, followed by a custom classifier for mask detection. The classifier was built by fine-tuning the MobileNetV2 architecture, achieving high accuracy and demonstrating a robust application of transfer learning for a critical public health challenge.

Evaluating Hybrid RAG and Graph RAG for Contextual Recommendations

Designed and implemented a sophisticated recommendation engine using a hybrid Retrieval-Augmented Generation (RAG) architecture. The system integrates traditional search with vector embeddings and is enhanced with a Graph RAG pipeline to leverage interconnected data, delivering highly context-aware recommendations. The entire workflow was orchestrated using Python with LangChain and Haystack, and its performance was rigorously validated for accuracy and relevance using the Fiddler AI Observability platform.

AI-Powered Sales Agent for Automated Lead Outreach

Architected a multi-modal AI agent that orchestrates complex lead engagement workflows for the Civix AI CRM. The system integrates automated calling (VoIP), messaging (SMS), and email APIs to execute targeted outreach campaigns. It utilizes LLMs to understand and generate human-like replies, effectively qualifying prospects and managing the initial stages of the sales funnel, demonstrating a practical application of agentic AI.

HelloGov AI: An Intelligent Automation Platform for Government Services

Engineered the AI core for the HelloGov platform, designed to automate U.S. passport submissions with high reliability. This involved creating an intelligent document processing system that leverages OCR and a fine-tuned Llama 3.2 model for data extraction and validation. A key innovation was a resilient Generative AI agent for human-like portal navigation, designed to operate securely at scale while minimizing detection and ensuring high first-time application approval rates.

Gesture-Based Human-Computer Interaction System for Application Control

Built a cross-platform gesture-controlled HCI system in Python that lets users launch and manage applications via customizable multi-finger gestures. Implemented a Tkinter GUI with persistent, device-specific user profiles stored in MongoDB and local JSON, demonstrating practical research in natural, gesture-driven interaction for improved usability and accessibility.

STANDARD TEST SCORES

IELTS (7.0, CEFR C1): Listening : 7.5 / Reading : 7.5 / Writing : 6 / Speaking : 6

GRE (331/340): Verbal Reasoning : 166 / Quantitative Reasoning : 165

Duolingo English Test (125/160): Speaking: 125 / Writing: 120 / Reading: 120 / Listening: 130

CERTIFICATIONS

- AWS Solutions Architect, Oct 25, 2024
- Oracle Certified Generative AI Professional, Jul 29, 2024
- Deloitte STEM Connect Virtual Experience Program, Mar 18, 2022
- Perform Data Science with Azure Databricks, Feb 18, 2022
- Microsoft Software Engineering Virtual Experience Program, Oct 19, 2020
- Commonwealth Bank Australia Careers in Technology, Sep 05, 2020
- Walmart Global Tech Advanced Software Engineering Virtual Experience, Jul 22, 2022
- AI Essentials for Healthcare Leaders Conference, Aug 30, 2025
- 3rd & 4th International Conference on Computing, Mathematics and Engineering Technologies, (iCoMET 2020 & iCoMET 2022)

LEADERSHIP & COMMUNITY INVOLVEMENT

- Organized university-wide thalassemia awareness campaigns and led digital skills workshops for students.
- Mentored junior and underprivileged students on academics, internships, and professional development.
- **Cultural Representative:** Represented Sukkur IBA University at the 15th South Asian Management Forum.
- **Sports Representative:** Represented Sukkur IBA University in badminton at a National Level Sports Festival.